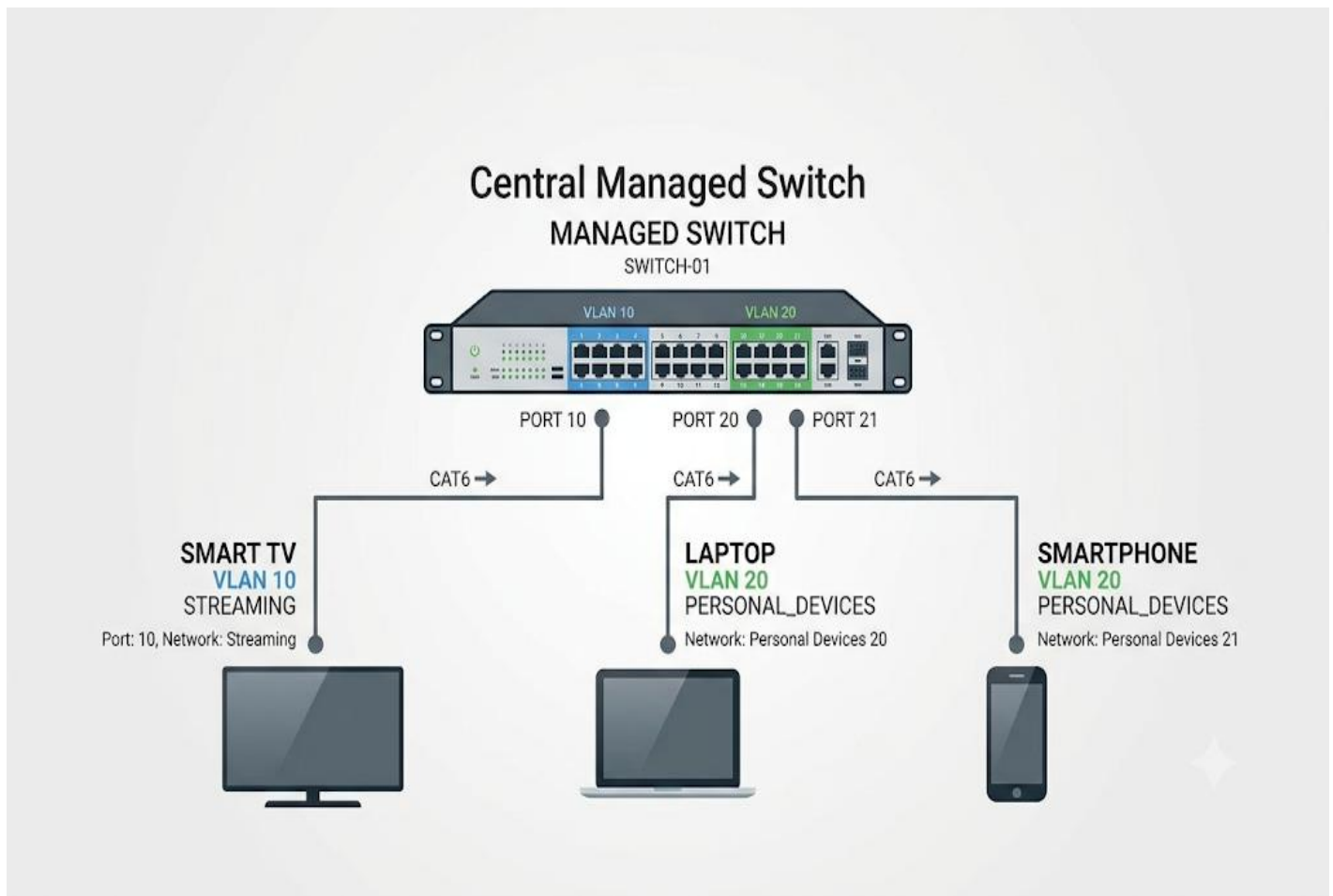


SWITCHES AND VLANS

1. IMAGE OF NETWORK DEVICE :



2. SETTING UP 2 GROUPS (VLANs) ON A SWITCH :

Step 1: Create the two groups

Plaintext

```
Switch> enable
```

```
Switch# configure terminal
```

! Create Group 10 for Streaming

```
Switch(config)# vlan 10
```

```
Switch(config-vlan)# name STREAMING
```

```
Switch(config-vlan)# exit
```

! Create Group 20 for Personal Devices

```
Switch(config)# vlan 20
```

```
Switch(config-vlan)# name PERSONAL
```

```
Switch(config-vlan)# exit
```

Step 2: Plug computers into their assigned groups

Plaintext

! Put PC1, PC2, and PC3 into Group 10 (Ports 1, 2, and 3)

```
Switch(config)# interface range fastEthernet 0/1 - 3
```

```
Switch(config-if-range)# switchport access vlan 10
```

```
Switch(config-if-range)# exit
```

! Put PC4 and PC5 into Group 20 (Ports 4 and 5)

```
Switch(config)# interface range fastEthernet 0/4 - 5
```

```
Switch(config-if-range)# switchport access vlan 20
```

```
Switch(config-if-range)# exit
```

Step 3: Test connection (Ping)

- **PC1 to PC2 (Both in VLAN 10): SUCCESS.** They are in the same group, so they can talk.
- **PC1 (VLAN 10) to PC4 (VLAN 20): FAILED (Request Timed Out).** They are in different groups, so the switch blocks them.

3. MOVING A PC TO A NEW GROUP :

IMAGINE MOVING A PC PLUGGED INTO **PORT 3** FROM **VLAN 10** TO **VLAN 20**.

BEFORE CHANGE

- **CHECK CURRENT SETUP:** TYPE SHOW VLAN BRIEF. PORT 3 IS CURRENTLY LISTED UNDER **VLAN 10**.
- **TEST:** PC3 CAN TALK TO PC 1 (VLAN 10).

THE COMMAND TO MOVE IT

PLAINTEXT

SWITCH# CONFIGURE TERMINAL

SWITCH(CONFIG)# INTERFACE FASTETHERNET 0/3

SWITCH(CONFIG-IF)# SWITCHPORT ACCESS VLAN 20

SWITCH(CONFIG-IF)# END

AFTER CHANGE

- **CHECK NEW SETUP:** TYPE SHOW VLAN BRIEF. PORT 3 NOW SHOWS UNDER **VLAN 20**.
- **TEST:** PC3 CAN NO LONGER TALK TO PC 1 (VLAN 10), BUT IT CAN NOW TALK TO PC4 (VLAN 20).

4. FAST SHORTCUT: ASSIGN 10 PORTS AT ONCE :

INSTEAD OF TYPING THE SAME COMMAND 10 SEPARATE TIMES FOR 10 PORTS, USE THE RANGE COMMAND TO SET PORTS 1 THROUGH 10 TO **VLAN 30** ALL AT ONCE.

PLAINTEXT

! 1. GO INTO SETUP MODE

CONFIGURE TERMINAL

! 2. CREATE VLAN 30

VLAN 30

NAME GUEST

EXIT

! 3. SELECT PORTS 1 THROUGH 10 ALL TOGETHER

INTERFACE RANGE FASTETHERNET 0/1 - 10

! 4. MOVE ALL 10 PORTS TO VLAN 30 INSTANTLY

```
SWITCHPORT MODE ACCESS  
SWITCHPORT ACCESS VLAN 30  
EXIT
```

```
! 5. SAVE YOUR WORK
```

```
END
```

```
WRITE MEMORY
```